

OPTIMAL DESIGN FOR INVERSE QUADRATIC POLYNOMIALS

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Summary: The four-parameter inverse quadratic polynomial model is commonly used for investigating the dependence of crop yields on amount of fertilizer applied and interest has focused on designing experiments so that the parameters of the model are estimated as precisely as possible. In particular, designs for which the determinant of the information matrix is a maximum, i.e. D_θ -optimal designs, have been computed numerically by Cobby, Chapman and Pike (1986) and shown to comprise four points of support with values in a geometric progression. In this article methods based on the theory of optimal experimental design are invoked to derive this result algebraically. Further it is shown that there are in fact two families of D_θ -optimal designs for the inverse quadratic polynomial model depending on the parameter values. Both of these families comprise exact designs, and explicit formulae for the support points of each are presented.

1. INTRODUCTION

The inverse quadratic polynomial model is defined by

$$(1.1) \quad y_i = \frac{x_i + \alpha}{\beta_0 + \beta_1(x_i + \alpha) + \beta_2(x_i + \alpha)^2} + \epsilon_i, \quad i = 1, \dots, n$$

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where y_i represents an observation taken at the corresponding experimental setting x_i with $0 < x_1 < \dots < x_n$, α , β_0 , β_1 and β_2 are unknown parameters such that $\beta_0, \beta_2 \geq 0$, $0 \leq \alpha < (\beta_0/\beta_2)^{\frac{1}{2}}$ and $-2(\beta_0\beta_2)^{\frac{1}{2}} < \beta_1 < 2(\beta_0\beta_2)^{\frac{1}{2}}$ and the terms ϵ_i ($i = 1, \dots, n$) are uncorrelated random components with mean 0 and constant variance σ^2 (Nelder (1966); Cobby, Chapman and Pike (1986)). The constraints imposed on the parameters ensure that the expected response function exhibits no discontinuities and has a single maximum at some nonnegative experimental setting. Model (1.1) is commonly used for investigating the dependence of crop yields on amount of fertilizer applied (Sparrow (1979)) and interest has focused on designing experiments, i.e. choosing the settings $x_1, \dots, x_n$, so that the unknown parameters are estimated precisely while at the same time some degrees of freedom are available for testing lack of fit. To appraise the efficiency of estimation for such designs, Cobby, Chapman and Pike (1986) computed D_θ -optimal designs, i.e. designs which maximize the determinant of the information matrix (Silvey (1980, p. 54)), for a range of fixed parameter values. They found in all cases considered that the D_θ -optimal designs comprise four points, one at zero and the other three in a geometric progression, and remarked that "The regularity of these results strongly suggests that there exists an underlying theoretical solution which has, so far, proved unobtainable." Such a solution is now presented here.

2. GENERAL APPROACH

We begin by introducing some results relating to optimal designs for nonlinear regression models with a single explanatory variable. Consider model (1.1) but with the i -th expected response defined by $\eta_i = \eta(x_i, \underline{\theta})$ where the experimental setting x_i is an element of a compact design space $\mathcal{X}$, $\underline{\theta}$ is a $p \times 1$ vector of unknown parameters and η is a known nonlinear response function continuously differentiable in $\underline{\theta}$. Suppose further that μ represents a design measure on $\mathcal{X}$ which puts probabilities μ_i at n distinct design points x_i , $i = 1, \dots, n$, respectively. Then μ is termed an

approximate design, the information matrix of μ for some parameter value $\underline{\theta}$ is defined by

$$M(\mu, \underline{\theta}) = \frac{1}{\sigma^2} \left[\sum_{i=1}^n \mu_i \frac{\partial \eta_i}{\partial \underline{\theta}_r} \frac{\partial \eta_i}{\partial \underline{\theta}_s} \right],$$

and the variance function at some experimental setting x is defined by

$$d(x, \mu, \underline{\theta}) = \left[\frac{\partial \eta}{\partial \underline{\theta}} \right]^T [M(\mu, \underline{\theta})]^{-1} \left[\frac{\partial \eta}{\partial \underline{\theta}} \right]$$

where $\frac{\partial \eta}{\partial \underline{\theta}}$ is a $p \times 1$ vector with the r -th element equal to $\frac{\partial \eta(x, \underline{\theta})}{\partial \underline{\theta}_r}$.

The D_θ -optimal design is that design for which the determinant of the information matrix $|M(\mu, \underline{\theta})|$ is a maximum and the Equivalence Theorem pertaining to such designs and proved by White (1973) and by Silvey (1980, p. 54) can be expressed as follows:

Theorem 1. A D_θ -optimal design $\mu_*(\underline{\theta})$ can be equivalently characterized by any of the three conditions

1. $\mu_*(\underline{\theta})$ maximizes $|M(\mu, \underline{\theta})|$,
2. $\mu_*(\underline{\theta})$ minimizes $\max_{x \in \mathcal{X}} d(x, \mu, \underline{\theta})$,
3. $\max_{x \in \mathcal{X}} d(x, \mu, \underline{\theta}) = p$.

Corollary. The variance function $d(x, \mu_*(\underline{\theta}), \underline{\theta})$ attains its maximum values at the support points of $\mu_*(\underline{\theta})$.

The method for constructing D_θ -optimal designs used in the present study can now be described within the above framework as follows.

1. Show that the variance function $d(x, \mu, \underline{\theta})$ for any design μ has at most p maxima on the design space $\mathcal{X}$. It then follows from the Corollary to Theorem 1 that the D_θ -optimal design cannot have

more than p points of support. At the same time this design must have at least p support points in order to ensure that its information matrix is nonsingular, so that overall we can conclude that the D_θ -optimal design is based on *exactly* p distinct points.

2. Observe now that for any design comprising p distinct points

$$|M(\mu, \underline{\theta})| = \frac{1}{(\sigma^2)^p} \left(\prod_{i=1}^p \mu_i \right) |X(\underline{\theta})|^2$$

where $X(\underline{\theta})$ is the $p \times p$ matrix with (i, r) -th element $\frac{\partial \eta_i}{\partial \theta_r}$. It thus follows immediately that the p -point D_θ -optimal design, for which this determinant is a maximum, puts equal probabilities $\mu_i = \frac{1}{p}$, $i = 1, \dots, p$, on the support points and is thus an exact design and further that these support points are the values of x_i , $i = 1, \dots, p$, for which the absolute value of $|X(\underline{\theta})|$, say $T(\underline{x}, \underline{\theta})$ where $\underline{x} = [x_1 \ x_2 \ \dots \ x_p]'$, is a maximum.

3. Maximize the function $T(\underline{x}, \underline{\theta})$ analytically with respect to $\underline{x}$ and hence derive explicit expressions for the support points of the exact D_θ -optimal design in terms of the parameters $\underline{\theta}$. It is interesting to note that the above strategy is very similar to that used by Fedorov (1972, pp. 83-94) and Silvey (1980, pp. 42-43) for constructing D -optimal designs for polynomial regression with a single variable.

3. D_θ -OPTIMAL DESIGNS FOR INVERSE QUADRATIC POLYNOMIALS

3.1 Location parameter α omitted

We begin by constructing D_θ -optimal designs for the inverse quadratic polynomial model with the parameter α omitted. In this case

$\eta(x, \underline{\theta}) = \frac{x}{\beta_0 + \beta_1 x + \beta_2 x^2}$ where $\underline{\theta} = [\beta_0 \ \beta_1 \ \beta_2]'$ and x belongs to the design space $\mathcal{X} = [0, \infty)$. The variance function $d(x, \mu, \underline{\theta})$, which is

nonnegative by definition, can be expressed as $\frac{x^2 P_4(x)}{(\beta_0 + \beta_1 x + \beta_2 x^2)^4}$ where

$P_4(x)$ is a polynomial in x of degree 4 and we firstly consider its form for $x \in \mathbb{R}$. In particular the value of $d(x, \mu, \underline{\theta})$ approaches zero as x approaches

plus or minus infinity and also setting $\frac{\partial d(x, \mu, \underline{\theta})}{\partial x} = 0$ yields a polynomial

equation in x of degree 7 with a root at $x = 0$ so that the function $d(x, \mu, \underline{\theta})$ has 7 stationary points on $\mathbb{R}$ including one with a value of zero at $x = 0$.

Thus $d(x, \mu, \underline{\theta})$ has at most 3 minima, including one at $x = 0$, and at most

4 maxima, of which at least one occurs at a strictly negative value of x , on

the domain $\mathbb{R}$. If we now restrict x to be an element of the design space

$\mathcal{X} = [0, \infty)$ then it is clear that $d(x, \mu, \underline{\theta})$ can have at most 3 maxima on

that space and hence, following the arguments developed in Section 2, that

the required D_θ -optimal design is an exact 3-point design with support

points given by the values of x_1, x_2 and x_3 for which $T(x, \underline{\theta})$ is a

maximum.

We now consider maximizing $T(x, \underline{\theta})$ analytically with respect to

x_1, x_2 and x_3 and in order to avoid any obvious ambiguities we assume that

$x_3 > x_2 > x_1 > 0$. It is then readily shown that

$$T(x, \underline{\theta}) = \frac{x_1 x_2 x_3 (x_3 - x_2)(x_3 - x_1)(x_2 - x_1)}{(\beta_0 + \beta_1 x_1 + \beta_2 x_1^2)^2 (\beta_0 + \beta_1 x_2 + \beta_2 x_2^2)^2 (\beta_0 + \beta_1 x_3 + \beta_2 x_3^2)^2}$$

and setting $\frac{\partial T(x, \underline{\theta})}{\partial x_i} = 0$, $i = 1, 2, 3$, and simplifying gives the three

equations:

$$(3.1a) \quad \beta_0[x_2x_3 - 2x_1(x_2+x_3) + 3x_1^2] + \beta_1x_1[x_1^2 - x_2x_3] - \beta_2x_1^2[3x_2x_3 - 2x_1(x_2+x_3) + x_1^2] = 0$$

$$(3.1b) \quad \beta_0[x_1x_3 - 2x_2(x_1+x_3) + 3x_2^2] + \beta_1x_2[x_2^2 - x_1x_3] - \beta_2x_2^2[3x_1x_3 - 2x_2(x_1+x_3) + x_2^2] = 0$$

$$(3.1c) \quad \beta_0[x_1x_2 - 2x_3(x_1+x_2) + 3x_3^2] + \beta_1x_3[x_3^2 - x_1x_2] - \beta_2x_3^2[3x_1x_2 - 2x_3(x_1+x_2) + x_3^2] = 0$$

Now solutions to these equations can be obtained by assuming that $x_1 = \frac{x_2}{r}$

and $x_3 = rx_2$ for some ratio r . Substituting into (3.1b) gives $x_2 = \pm \sqrt{\frac{\beta_0}{\beta_2}}$ and substituting into (3.1a), or equivalently into (3.1c), yields the quintic polynomial

$$(3.2) \quad r^5 - (\gamma + 2)r^4 - 5r^3 + 5r^2 + (\gamma + 2)r - 1 = 0$$

where $\gamma = \frac{\beta_1}{\sqrt{\beta_0\beta_2}}$. To satisfy the condition $x_3 > x_2 > x_1 > 0$ we select

$x_2 = \sqrt{\frac{\beta_0}{\beta_2}}$ and note that r must be strictly greater than 1. Now $r = 1$ is a solution to equation (3.2) and factorizing out this root gives the quartic polynomial equation

$$r^4 - (\gamma + 1)r^3 - (\gamma + 6)r^2 - (\gamma + 1)r + 1 = 0.$$

This is a reciprocal equation which is readily solved (Heading (1963, pp. 34-35)) and it then follows from the restrictions placed on the parameters β_0, β_1 and β_2 by the model formulation that only one of the solutions for r

so obtained, viz. $\rho = \frac{\delta + \sqrt{\delta^2 - 4}}{2}$ where $\delta = \frac{\gamma + 1 + \sqrt{\gamma^2 + 6\gamma + 33}}{2}$, is greater than 1.

Thus the function $T(x, \theta)$ has a maximum at $\underline{x}^* = \left[\frac{1}{\rho} \sqrt{\frac{\beta_0}{\beta_2}}, \sqrt{\frac{\beta_0}{\beta_2}}, \rho \sqrt{\frac{\beta_0}{\beta_2}} \right]^T$, and further it is readily shown that this maximum value, $T(\underline{x}^*, \theta)$, is given by

$$\frac{(\rho - 1)^3(\rho + 1)\rho^2}{[2\sqrt{\beta_0\beta_2} + \beta_1]^2[(1 + \rho^2)\sqrt{\beta_0\beta_2} + \rho\beta_1]^4}.$$

Overall, therefore, the D_θ -optimal design for the inverse quadratic polynomial with the parameter α omitted is an exact 3-point design with

support points at $\frac{1}{\rho} \sqrt{\frac{\beta_0}{\beta_2}}$, $\sqrt{\frac{\beta_0}{\beta_2}}$ and $\rho \sqrt{\frac{\beta_0}{\beta_2}}$. This result is illustrated with a numerical example, taking $\beta_0 = \beta_2 = 1$ and $\beta_1 = -1.1$. The values of β_0 and β_2 were chosen in accord with those investigated by Cobby, Chapman and Pike (1986) and the value of β_1 was selected so that the salient features of the variance function are displayed to best advantage. For these parameter values the D_θ -optimal design puts equal probabilities on the three points 0.476, 1.000 and 2.101 and has a $T(\underline{x}, \theta)$ value of 0.2433. A plot of $d(x, \mu_*(\theta), \theta)$ against x for this design is shown in figure 1. The form of this graph highlights the features of the variance function discussed above and also confirms the optimality of the design over the design space $\mathcal{X} = [0, \infty)$.

3.2 Location parameter α included

We now consider the inverse quadratic polynomial model with the parameter α included, i.e. model (1.1). Then the variance function $d(x, \mu, \theta)$, where μ is any design measure and $\theta = [\alpha\beta_0\beta_1\beta_2]'$, is nonnegative definite by

definition and has the form $\frac{P_6(x)}{[\beta_0 + \beta_1(x + \alpha) + \beta_2(x + \alpha)^2]^4}$ where $P_6(x)$ is a polynomial in x of degree 6. Thus $d(x, \mu, \theta)$ has limiting values of zero as

x approaches plus and minus infinity and also setting $\frac{\partial d(x, \mu, \theta)}{\partial x} = 0$

gives a polynomial equation in x of degree 7 so that $d(x, \mu, \theta)$ has 7 stationary points on $\mathbb{R}$. Overall therefore the function $d(x, \mu, \theta)$ has at most 4 maxima on $\mathbb{R}$ and hence on the design space $\mathcal{X} = [0, \infty)$ and thus it follows from section 2 that the D_θ -optimal design is an exact 4-point design with support points at those values of $x_i, i = 1, \dots, 4$, for which the function $T(x, \theta)$ is a maximum.

We firstly consider maximizing $T(x, \theta)$ with respect to x where $-\infty < x_1 < x_2 < x_3 < x_4 < \infty$ and thus the design points are not constrained to lie in the design space $\mathcal{X} = [0, \infty)$. It is readily shown that

$$T(x, \theta) = \frac{\beta_0(z_4 - z_3)(z_4 - z_2)(z_4 - z_1)(z_3 - z_2)(z_3 - z_1)(z_2 - z_1)}{(\beta_0 + \beta_1 z_1 + \beta_2 z_1^2)^2 (\beta_0 + \beta_1 z_2 + \beta_2 z_2^2)^2 (\beta_0 + \beta_1 z_3 + \beta_2 z_3^2)^2 (\beta_0 + \beta_1 z_4 + \beta_2 z_4^2)^2}$$

where $z_i = x_i + \alpha, i = 1, \dots, 4$. Setting $\frac{\partial T(x, \theta)}{\partial x_i} = 0$ for $i = 1, \dots, 4$, gives the equations

$$(\beta_0 + \beta_1 z_i + \beta_2 z_i^2) \left[\sum_{\substack{j \neq i \\ j=1, \dots, 4}} \frac{1}{(z_i - z_j)} \right] - 2(\beta_1 + 2\beta_2 z_i) = 0, \quad i = 1, \dots, 4$$

and following Fedorov (1972, pp. 90-91), these equations can be re-expressed as

$$(3.3) \quad (\beta_0 + \beta_1 z_i + \beta_2 z_i^2) f''(z_i) - 4(\beta_1 + 2\beta_2 z_i) f'(z_i) = 0, \quad i = 1, \dots, 4$$

where the function $f(z)$ is defined as $(z - z_1)(z - z_2)(z - z_3)(z - z_4)$ and $f'(z)$ and $f''(z)$ are the first- and second-order derivatives of $f(z)$ respectively. Thus the points $z_i, i = 1, \dots, 4$, which satisfy the equations (3.3) are the roots of the quartic polynomials $(\beta_0 + \beta_1 z + \beta_2 z^2) f''(z) - 4(\beta_1 + 2\beta_2 z) f'(z)$ and $f(z)$, and thus $f(z)$ satisfies the second-order differential equation

$$(\beta_0 + \beta_1 z + \beta_2 z^2) \frac{d^2 y}{dz^2} - 4(\beta_1 + 2\beta_2 z) \frac{dy}{dz} + 20\beta_2 y = 0$$

where y is some function of z . Now changing the variable z in this

equation to $w = \frac{2\beta_2 z + \beta_1}{\sqrt{4\beta_0\beta_2 - \beta_1^2}}$ gives the differential equation

$$(w^2 + 1) \frac{d^2 y}{dw^2} - 8w \frac{dy}{dw} + 20y = 0$$

which has a general series solution (Boyce & DiPrima (1986, pp. 194–215)) expressible in closed form as

$$y = a_0(1 - 10w^2 + 5w^4) + a_1(w - 2w^3 + \frac{1}{5}w^5)$$

where a_0 and a_1 are arbitrary constants. Then, since the required solution is a quartic polynomial in z and hence in w , it follows that this solution is obtained by setting $a_1 = 0$ and is determined up to a constant multiplier by

$$5w^4 - 10w^2 + 1.$$

Thus the points x_i , $i = 1, \dots, 4$, satisfying the equations $\frac{\partial T(\underline{x}, \underline{\theta})}{\partial x_i} = 0$ are obtained from the roots of this latter polynomial and can be written explicitly as

$$(3.4) \quad \left\{ \begin{array}{l} x_1 = -\alpha - \frac{\beta_1}{2\beta_2} - \frac{1}{2\beta_2} \sqrt{(1 + \frac{2}{\sqrt{5}})(4\beta_0\beta_2 - \beta_1^2)} \\ x_2 = -\alpha - \frac{\beta_1}{2\beta_2} - \frac{1}{2\beta_2} \sqrt{(1 - \frac{2}{\sqrt{5}})(4\beta_0\beta_2 - \beta_1^2)} \\ x_3 = -\alpha - \frac{\beta_1}{2\beta_2} + \frac{1}{2\beta_2} \sqrt{(1 - \frac{2}{\sqrt{5}})(4\beta_0\beta_2 - \beta_1^2)} \\ x_4 = -\alpha - \frac{\beta_1}{2\beta_2} + \frac{1}{2\beta_2} \sqrt{(1 + \frac{2}{\sqrt{5}})(4\beta_0\beta_2 - \beta_1^2)} \end{array} \right.$$

Further it readily follows that the maximum value of $T(\underline{x}, \underline{\theta})$ attained at these points has the simple form

$$\frac{25\sqrt{5} \beta_0 \beta_2^2}{(4\beta_0\beta_2 - \beta_1^2)^5}.$$

Overall therefore the D_θ -optimal design for model (1.1) with the extended design space $\mathbb{R}$ puts equal probabilities on points of support given by (3.4).

We are concerned in this study, however, with the design space $\mathcal{X} = [0, \infty)$ so that the points specified by (3.4) only give the support points for the required D_θ -optimal design provided x_1 is greater than or equal to zero and hence, by rearrangement of this inequality, provided the condition

$$(3.5) \quad \frac{\beta_1 + 2\beta_2\alpha}{\sqrt{\beta_2(\beta_0 + \beta_1\alpha + \beta_2\alpha^2)}} \leq -\frac{(1 + \sqrt{5})}{2}$$

holds. Otherwise it is clear from the general form of the variance function $d(x, \mu, \underline{\theta})$ described above that this function can only have the requisite 4 maxima on $[0, \infty)$ provided one occurs at zero. Thus if condition (3.5) does not hold, then the required D_θ -optimal design necessarily has a support point at $x_1 = 0$ and also at the values of x_2, x_3 and x_4 which maximize $T(\underline{x}, \underline{\theta})$ with x_1 set equal to zero, i.e. which maximize

$$T'(\underline{x}, \underline{\theta}) = \frac{\beta_0 x_2 x_3 x_4 (x_4 - x_3)(x_4 - x_2)(x_3 - x_2)}{(\beta_0 + \beta_1\alpha + \beta_2\alpha^2)^2 (\beta_0 + \beta_1 x_2 + \beta_2 x_2^2)^2 (\beta_0 + \beta_1 x_3 + \beta_2 x_3^2)^2 (\beta_0 + \beta_1 x_4 + \beta_2 x_4^2)^2}$$

where $x_4 > x_3 > x_2 > 0$. Now this optimization problem is, with certain obvious modifications, the same as that described in section 3.1 and thus it follows that the required D_θ -optimal design is the exact 4-point design with support points given by

$$(3.6) \quad \left\{ \begin{array}{l} x_1 = 0 \\ x_2 = \frac{1}{\rho_\alpha} \sqrt{\frac{\beta_0 + \beta_1 \alpha + \beta_2 \alpha^2}{\beta_2}} \\ x_3 = \sqrt{\frac{\beta_0 + \beta_1 \alpha + \beta_2 \alpha^2}{\beta_2}} \\ x_4 = \rho_\alpha \sqrt{\frac{\beta_0 + \beta_1 \alpha + \beta_2 \alpha^2}{\beta_2}} \end{array} \right.$$

where $\rho_\alpha = \frac{\delta_\alpha + \sqrt{\delta_\alpha^2 - 4}}{2}$, $\delta_\alpha = \frac{\gamma_\alpha + 1 + \sqrt{\gamma_\alpha^2 + 6\gamma_\alpha + 33}}{2}$ and

$$\gamma_\alpha = \frac{\beta_1 + 2\beta_2 \alpha}{\sqrt{\beta_2(\beta_0 + \beta_1 \alpha + \beta_2 \alpha^2)}}, \quad \text{and further that the maximum value of}$$

$T'(\underline{x}, \underline{\theta})$ is given by

$$\frac{\beta_0(\rho_\alpha - 1)^3(\rho_\alpha + 1)\rho_\alpha^2}{\beta_{0\alpha}^2 [2\sqrt{\beta_{0\alpha}\beta_2} + \beta_{1\alpha}]^2 [(1 + \rho_\alpha^2)\sqrt{\beta_{0\alpha}\beta_2} + \rho_\alpha\beta_{1\alpha}]^4},$$

where $\beta_{0\alpha} = \beta_0 + \beta_1 \alpha + \beta_2 \alpha^2$ and $\beta_{1\alpha} = \beta_1 + 2\beta_2 \alpha$.

It is interesting to note that the values of the support points of the D_θ -optimal designs derived here exhibit certain patterns. In particular the points defined by (3.4) can be expressed in relation to a geometric sequence

$$[\frac{a}{\lambda}, a, a\lambda] \quad \text{as} \quad x_1, x_1 + \frac{a}{\lambda}, x_1 + a, x_1 + \lambda a, \quad \text{where} \quad \lambda = \frac{\sqrt{5} + 1}{2} \quad \text{and}$$

$$a = (\sqrt{5} + 3) \sqrt{1 - \frac{2}{\sqrt{5}}} \sqrt{\frac{4\beta_0\beta_2 - \beta_1^2}{2\beta_2}} \quad \text{and also, since} \quad \lambda^2 - \lambda - 1 = 0, \quad \text{they}$$

satisfy the relation $x_2 - x_1 = x_4 - x_3$. More obviously the points x_2 , x_3 and x_4 specified in (3.6) have values in a geometric progression. It is also worth noting that the sets of points given by (3.4) and by (3.6) depend on the parameters of the model only through the location parameter α and the ratios $\frac{\beta_0}{\beta_2}$ and $\frac{\beta_1}{\beta_2}$ (cf. Cobby, Chapman and Pike (1986)). Indeed changing

the value of α for fixed $\frac{\beta_0}{\beta_2}$ and $\frac{\beta_1}{\beta_2}$ induces a change in location of the x_i -values, $i = 1, \dots, 4$, given by expression (3.4) and the corresponding maximum value of $T(\underline{x}, \underline{\theta})$ remains unchanged. The sensitivity of the points defined by (3.6) to the parameter α involves the term ρ_α and is more complicated, however.

We now illustrate the above results by means of two numerical examples. Again following Cobby, Chapman and Pike (1986), we adopt the parameter values $\alpha = 0,1$ and $\beta_0 = \beta_2 = 1$. Then for $\beta_1 = -1,8$ condition (3.5) holds and the D_θ -optimal design is the exact 4-point design with support points given by (3.4) as 0,200, 0,658, 0,942 and 1,400 and a $T(\underline{x}, \underline{\theta})$ value of 220,4738. In contrast for $\beta_1 = -1,5$ condition (3.5) does *not* hold and the D_θ -optimal design puts equal probabilities on support points specified by (3.6) as 0, 0,507, 0,927 and 1,696 and a $T'(\underline{x}, \underline{\theta})$ value of 3,1815. The variance function plots for these two designs are shown in figure 2(a) and figure 3 respectively and confirm the D_θ -optimality. For $\beta_1 = -1,8$ it was also interesting to consider the design with support points specified by (3.6) at 0, 0,621, 0,911 and 1,337 and a $T'(\underline{x}, \underline{\theta})$ value of 210,2966. The design is not D_θ -optimal, as evidenced by the variance function plot shown in figure 2(b), and in fact has a % efficiency (Cobby, Chapman and Pike (1986)) of 97,7. Both of the D_θ -optimal designs derived here were checked numerically using a quasi-Newton optimization routine from the module OPTMUM of the programming language GAUSS. It was interesting to observe that for the case where $\beta_1 = -1,8$ the convergence properties of the routine were poor; this may well be ascribed to the presence of a stationary point of the restricted function at 0, 0,621, 0,911 and 1,337 'close' to the

optimum of the unrestricted function at 0,200, 0,658, 0,942 and 1,400 or, following a suggestion of Chaloner and Larntz (1989), to the relative flatness of the variance function plot. In contrast for $\beta_1 = -1,5$ convergence to the D_θ -optimal design was straightforward.

4. DISCUSSION

The main aim of this paper has been to derive D_θ -optimal designs for the four-parameter inverse quadratic polynomial model algebraically and in so doing to explain the geometric pattern in the design points found numerically by Cobby, Chapman and Pike (1986). In particular it has been shown that there are two families of D_θ -optimal designs for this model depending on the values of the parameters. Both of these families comprise exact designs and explicit formulae for the support points are given in (3.4) and in (3.6). The latter expressions are useful firstly in that they provide experimenters with a quick and easy way of finding D_θ -optimal designs for the inverse quadratic polynomial and secondly in that they show clearly the underlying geometric patterns and also the dependence of the support points on the location parameter α and the ratios $\frac{\beta_0}{\beta_2}$ and $\frac{\beta_1}{\beta_2}$. The optimal design with support points specified by (3.4) relates to parameter values which comply with condition (3.5) and is of particular interest in that it was not identified in the earlier numerical studies.

The example discussed in this paper illustrates certain techniques that can be used to construct optimal designs in more general situations. In particular if it can be shown that the variance function of a nonlinear model has at most p maxima on the specified design space, then it follows that the D_θ -optimal design is an exact p -point design and the problem of maximizing the determinant of the information matrix, $|M(\mu, \theta)|$, reduces to that of maximizing the absolute value of $|X(\theta)|$. Very often, as in the present case, the determinant $|X(\theta)|$ can be written down in a simple explicit form and standard methods of calculus used for maximization. It should be noted, however, that this strategy does not always work; in particular it is not

always possible to show that the variance function has at most p maxima on the design space.

Finally this work has a number of obvious extensions and in particular it would be interesting to consider other optimal design criteria for the inverse quadratic polynomial model. For example it would be practically meaningful to construct designs which minimize the variance of the nonlinear function describing the maximum of the expected response for a fixed value of θ and, more generally, it would be appealing to examine optimal Bayesian designs in which prior information about the unknown parameters is incorporated into suitable criteria (Chaloner and Larntz, 1989). However, these ideas are beyond the scope of the present paper.

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REFERENCES

- BOYCE, W.E. & DIPRIMA, R.C. (1986). *Elementary differential equations and boundary value problems*. 4th edition. John Wiley, New York.
- CHALONER, K. & LARNTZ, K. (1989). Optimal Bayesian design applied to logistic regression experiments. *Journal of Statistical Planning and Inference*, **21**, 191–208.
- COBBY, J.M., CHAPMAN, P.F. & PIKE, D.J. (1986). Design of experiments for estimating inverse quadratic polynomial responses. *Biometrics*, **42**, 659–664.
- HEADING, J. (1963). *Mathematical methods in science and engineering*. Edward Arnold, London.
- NELDER, J.A. (1966). Inverse polynomials, a useful group of multi-factor response functions. *Biometrics*, **22**, 128–141.
- SILVEY, S.D. (1980). *Optimal design*. Chapman and Hall, London.

SPARROW, P.E. (1979). The comparison of five response curves for representing the relationship between the annual dry-matter yield of grass herbage and fertilizer nitrogen. *Journal of Agricultural Science, Cambridge*, **93**, 513-520.

WHITE, L.V. (1973). An extension of the General Equivalence Theorem to nonlinear models. *Biometrika*, **60**, 345-348.

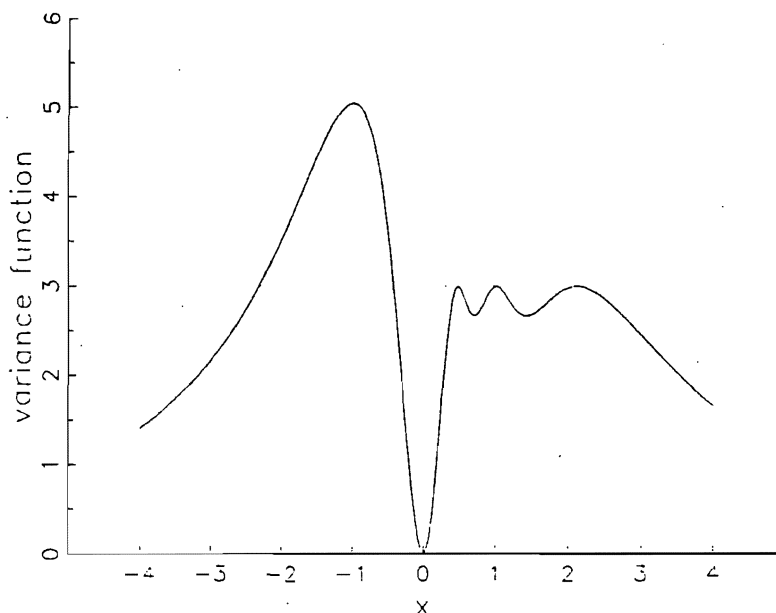


Figure 1. Variance function $d(x, \mu_*(\theta), \theta)$ plotted against x for the D_θ -optimal design for the inverse quadratic polynomial model with the parameter α omitted and with $\beta_0 = \beta_2 = 1$ and $\beta_1 = -1, 1$.

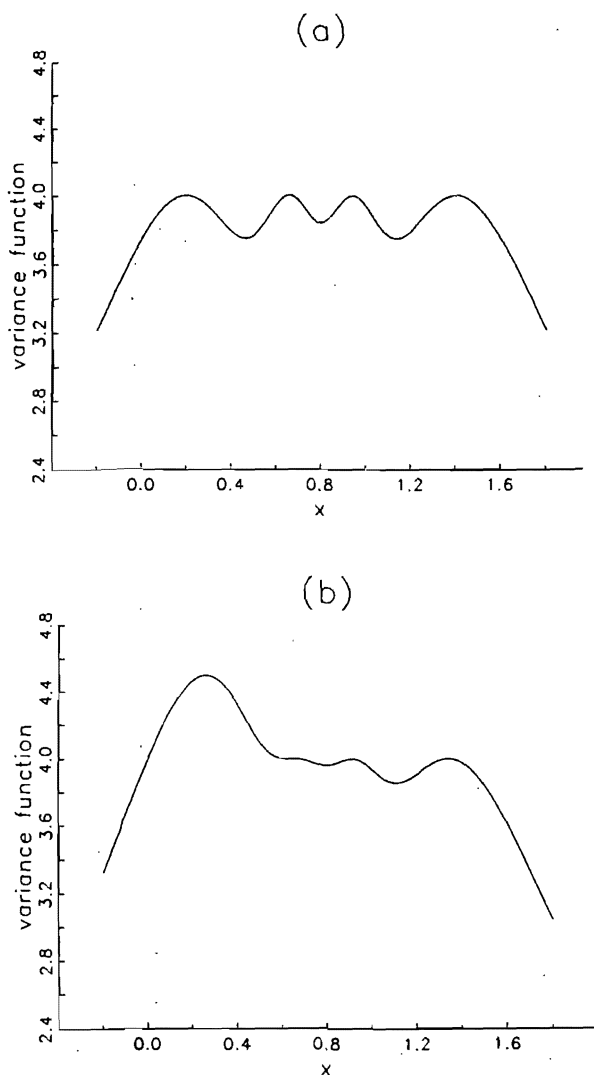


Figure 2. Variance function $d(x, \mu, \underline{\theta})$ plotted against x for the inverse quadratic polynomial model with $\alpha = 0, 1$, $\beta_0 = \beta_2 = 1$ and $\beta_1 = -1, 8$.
 (a) D_{θ} -optimal design. (b) Near-optimal design with $x_1 = 0$.

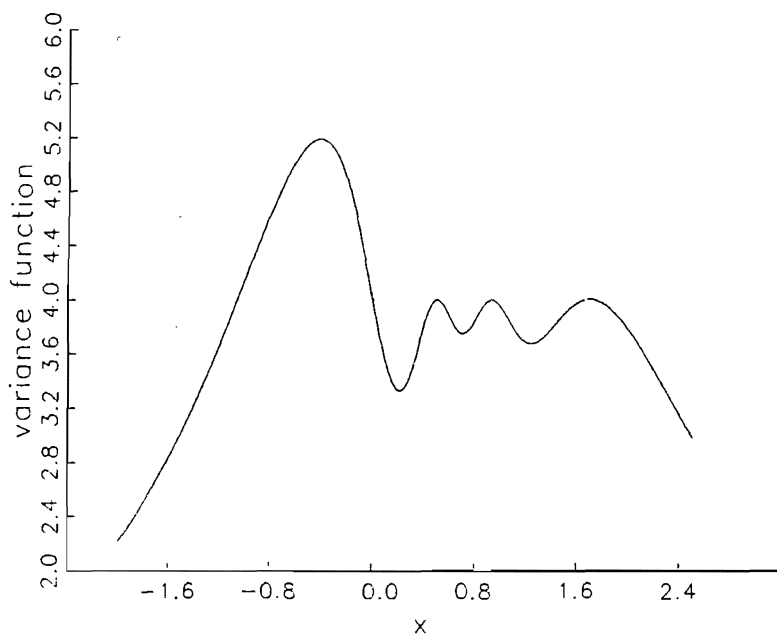


Figure 3. Variance function $d(x, \mu_*(\underline{\theta}), \underline{\theta})$ plotted against x for the D_{θ} -optimal design for the inverse quadratic polynomial model with $\alpha = 0, 1$, $\beta_0 = \beta_2 = 1$ and $\beta_1 = -1, 5$.